



DEPARTMENT OF CHEMICAL ENGINEERING

NATIONAL INSTITUTE OF TECHNOLOGY SRINAGAR

Bioresource Technology (CET 465)

Home Assignment plus Question Bank-7

S. No.	Questions	COs
1.	Discuss the significance of pretreatment in bioresource technology and explain the basics of hydrolysis in pretreatment.	CO3
2.	Summarize the pre-treatment methods by a flow chart diagram and briefly discuss them without missing any important point.	CO3
3.	Discuss the structure and composition of various LCBs	CO2
4.	When wood flour is subjected to ultrasonication, how is the disintegration of various components of LCB effected? Sketch the diagram and show it.	CO3
5.	Write short notes on chemical, physical, physicochemical and biological methods of pretreatments.	CO3
6.	How is the position of cellulose, hemicellulose and lignin effected in enzyme pretreatment process.	CO3
7.	Explain biomaterials and cite at least ten examples.	CO1
8.	Write a short note on the history of biomaterials.	CO1
9.	Discuss the applications of biomaterials.	CO2
10.	Define biocompatibility and discuss the requirement for implantation.	CO1
11.	Describe the characteristics and properties of biomaterials.	CO2
12.	How is the selection of biomaterials made, discuss?	CO3
13.	Discuss the most common classification of biomaterials.	CO2
14.	What is biopulping? Discuss its significance.	CO3
15.	Describe the concerns of conventional pulping process.	CO1
16.	Give an overview of the steps involved in biopulping process with the help of a simple diagram.	CO3
17.	List the white rot fungi species used for biopulping.	CO2
18.	Write a brief note on the constraints of biopulping.	CO1
CO1:	Fundamental understanding of the bioresources and its applications for attainment of social objectives (energy, environment, product, sustainability).	
CO2:	Acquire knowledge with respect to the properties of the bioresources and the conversion technologies.	
CO3:	Exhibiting knowledge of the systems used for bioresources and bioresource technology.	
CO4:	Understanding about analysis of data and their applications in design of the systems and development of the bioprocess.	